



COMPUTER SECURITY

2025

ORAL & PRACTICAL BANK (Solved)



SENIOR
25

BY AB & AE & AM & MR

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@qahwa_is



ORAL BANK

Choose the correct answer:

1.	The need for <u>keeping information secret</u> .			
	a) Confidentiality	b) Integrity	c) Availability	d) None
2.	Trustworthiness of data and resources.			
	a) Confidentiality	b) Integrity	c) Availability	d) None
3.	The ability to use the information or resource			
	a) Confidentiality	b) Integrity	c) Availability	d) None
4.	Timely response is an expectation of _____ security goal			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
5.	_____ means that some unauthorized party (person, program) <u>has gained access to data</u>			
	a) Interception	b) Modification	c) Interruption	d) Fabrication
6.	_____ an asset of the system <i>becomes lost, unavailable or unusable</i>			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
7.	_____ means that some unauthorized party (person, program) has <u><i>gained access to an asset</i></u>			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
8.	_____ an additional computation, or modifying data being transmitted by the network.			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
9.	_____ counterfeit objects on a computing system. (adding records to an existing data base or insertion of spurious transactions to a network communication system.			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
10.	Cryptography is used as a defense technique against the following attacks except _____			
	a) Modification	b) Interception	c) Interruption	d) Fabrication
11.	Erasure of program or data or malfunction of an OS file manager is considered _____			
	a) Modification	b) Interception	c) Interruption	d) Fabrication
12.	Flooding is an attack of _____			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
13.	Spoofing is an attack of _____			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
14.	_____ is when <u>one host pretends to be another</u>			
	a) Eavesdrop	b) Spoofing	c) Wiretap	d) Session Hijacking
15.	A sender <u><i>should not be able falsely to deny later that he/she sent a message</i></u>			
	a) Integrity	b) Confidentially	c) Privacy	d) Non-repudiation



16.	Program controls include:							
	a)	Peer review	b)	Proof of program correctness	c)	Independent testing	d)	All
17.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
18.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
19.	Eavesdropping and impersonate are attacks of _____ Respectively							
	a)	Availability , Confidentiality	b)	Availability , Integrity	c)	Confidentiality , Integrity	d)	Integrity , Confidentiality
20.	Choose the correct sentence?							
	a)	Erasure of a program is considered an interruption threat						
	b)	TOCTOU is considered an attack of availability						
	c)	Flooding are attacks of confidentiality						
	d)	Error is a mistake in interpreting a syntax error in a piece of code						
21.	_____ Consuming processor capacity, system memory, disk space							
	a)	Bacteria	b)	Resident	c)	Virus	d)	Worm
22.	_____ Attaches itself to program and propagates copies of itself to other programs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
23.	_____ Contains unexpected, additional functionality							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
24.	_____ Triggers action when condition occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
25.	_____ Triggers action when specified time occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
26.	_____ Allows unauthorized access to functionality							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
27.	_____ Propagates copies of itself through a network							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
28.	_____ Replicates itself without limit to exhaust resources							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
29.	_____ is a class of malicious code that "detonates" or goes off when a specified condition occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
30.	The _____ is a code embedded in program that is set to explode when certain condition is/are met							
	a)	Trojan house	b)	Trapdoor	c)	Worm	d)	Logical bomb



31.	_____ replicates a copy of itself <u>to other systems</u> to execute a malicious code			
	a) Virus	b) Worm	c) Trojan horse	d) Trapdoor
32.	_____ is a program that spreads copies of itself through a network			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
33.	_____ is a feature in a program by which someone can access the program <u>other than by the obvious, direct call</u> , perhaps with special privileges			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
34.	_____ virus runs when its attached program executes and terminates when its attached program ends			
	a) Transient	b) Resident	c) Virus	d) Worm
35.	_____ virus locates itself in memory so that it can <u>remain active</u> even after its attached program ends			
	a) Transient	b) Resident	c) Virus	d) Worm
36.	A type of virus that executes only once , spreading its infection and causing its effect in that one execution .			
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus
37.	A kind of malicious software runs when a computer is started , it changes control on the OS instructions			
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus
38.	A type of virus which is active as long as the computer running			
	a) Memory resident virus	b) Brain virus	c) Polymorphic virus	d) Logic bomb
39.	Virus is active as long as the computer running			
	a) Memory resident	b) Brain virus	c) Polymorphic virus	d) Trojan horse
40.	Viruses resident in application programs and libraries are categorized as:			
	a) One time execution virus	b) Boot sector virus	c) Other homes of virus	d) Memory resident
41.	A program virus attaches itself to a program ; then, whenever the program is run, the virus is activated. This kind of attachment is usually known as _____			
	a) integrated virus	b) Surrounded virus	c) Appended virus	d) replacement virus
42.	controls against program threats include the following actions <u>except</u> _____			
	a) peer review process	b) Independent testing	c) Proof of correctness	d) Decapsulation of different process
43.	Each virus has a signature that could be used to detect its existence, such as:			
	a) Its location	b) its execution performance	c) its replication technique	d) All
44.	What is a weakness of Caesar cipher?			
	a) Vulnerable to brute-force attacks	b) Vulnerable to frequency analysis	c) limited key space	d) all



45.	What is true about Steganography?							
	a)	it is used to encrypt images and send it in unreadable forms						
	b)	The encrypted file has smaller size compared with the original one						
	c)	it can be broken by brute-force attack						
	d)	it uses images to hide important data in LSB of each image's pixel						
46.	Steganography uses _____ in graphic image to hide important data							
	a)	LSB	b)	HSB	c)	MSB	d)	NSB
47.	_____ is a mistake in <i>interpreting a requirement</i> or a syntax error in a piece of code							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
48.	_____ is a human mistake in performing some software activity that <u>may lead to a fault</u> in computer program							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
49.	Validation error , domain error , and boundary condition violation are types of _____							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
50.	A _____ <u>may cause a failure</u> (which is a departure from the system's required behavior).							
	a)	Bug	b)	Error	c)	fault	d)	failure
51.	A _____ is an inside view of the system , <u>seen by the developers</u> , whereas a _____ is an outside view <u>seen by the user</u>							
	a)	Failure, fault	b)	Fault, Failure	c)	Error, Bug	d)	Fault, Flaws
52.	What is true about program <u>flaws</u> ?							
	a)	It could be validation error or domain error						
	b)	It could lead to failure						
	c)	It could be malicious or non malicious						
	d)	All are true						
53.	Consist of replacing the valid source and/or destination IP address with false ones							
	a)	Smurf attack	b)	Spoofing attack	c)	DoS	d)	Sniffing attack
54.	For exchanging symmetric key among 10 user , we need _____ exchange operation							
	$n = 10, \frac{n \times n - 1}{2} = \frac{10 \times 9}{2} = 45$							
	a)	45	b)	100	c)	90	d)	1000
55.	Authentication based on <u>Something the individual has</u> as:							
	a)	ID	b)	Password	c)	ATM card	d)	PIN
56.	Authentication based on <u>Something the individual knows</u> as:							
	a)	PIN	b)	Password	c)	ATM card	d)	A & B
57.	Authentication based on <u>Something the individual is</u> as:							
	a)	Voice patterns	b)	Handwriting	c)	Fingerprint	d)	All



58.	Password and PIN are examples for user authentication of something _____			
	a) The user has	b) The user is	c) The user knows	d) The user
59.	What is the Row Transposition Cipher?			
	a) A substitution cipher that rearranges the order of letters in a message			
	b) A cipher that replaces each letter in a message with a numerical value			
	c) A cipher that applies a mathematical formula to each character in a message			
	d) A cipher that rearranges the order of words in a message			
60.	Which of the following is a common technique used with transposition cipher?			
	a) Rail fence	b) Caesar	c) Vigenere	d) Playfair
61.	The mechanisms used to confirm a user's identity is known as:			
	a) Identification	b) Authentication	c) Authorization	d) Access limitation
62.	Each virus has a signature that could be used to detect its existence, such as:			
	a) Location	b) Execution performance	c) Spreading	d) All are true
63.	In _____, Users interact with directory to obtain any desired public key securely			
	a) Certificate authority	b) Public Key authority	c) Public announcement	d) Public Discovery
64.	_____ allow key exchange without real-time access to public-key authority			
	a) Certificate authority	b) Public key authority	c) Public announcement	d) Public directory
65.	Example for classical substitution encryption			
	a) Playfair	b) DES	c) 3DES	d) AES
66.	Triple-DES with two keys is defined by $C = EK_3(DK_2(EK_1(P)))$, if K_1 equal K_3 , the triple-DES can be considered هنا هو بيقولك ان k_1 يساوي k_3 ، وأول خطوه حصلت انه شفر ال P ب k_1 وبعدين تاني خطوة فك تشفيرها ب k_2 بس ده مش هيرجع لك P الاصلية لان k_1 مش هي k_2 فدي هتعتبر خطوة تشفير تانية، وبعدين قام مشفر ب k_3 فكذا حصل تشفير ثلاث مرات يبقى ده triple DES			
	a) Double DES	b) Triple-DES	c) Single-DES	d) Useless encryption
67.	Triple-DES with two keys is defined by $C = EK_1(DK_2(EK_1(P)))$, if K_1 equals K_2 , the triple-DES will be هنا هو بيقولك ان k_1 يساوي k_2 ، وأول خطوه حصلت انه شفر ال P ب k_1 وبعدين تاني خطوة فك تشفيرها ب k_2 فهيرجع ال P الأصلية عادي يعني اول خطوه اتلغت بتاني خطوة، وبعدين رجع شفر ب k_1 فيبقى كأنه عمل تشفير مره واحده بس فوتهبقى single DES			
	a) Double DES	b) Triple-DES	c) Single-DES	d) Useless encryption
68.	Which of the following is a good description for Triple-DES ?			
	a) $C = EK_1(DK_2(EK_1(P)))$	b) $C = EK_2(DK_1(EK_1(P)))$	c) $C = EK_3(EK_2(EK_1(P)))$	d) $C = DK_3(DK_2(DK_1(P)))$
69.	DES encrypt data in blocks of size _____, using a key of length _____			
	a) 64-bit , 32-bit	b) 32-bit, 32-bit	c) 32-bit , 56-bit	d) 64-bit , 56-bit



70.	What is the number of rounds in <u>single DES</u> ?							
	a)	64	b)	56	c)	48	d)	16
71.	What is the block size of <u>triple DES</u> ?							
	a)	64	b)	128	c)	256	d)	512
72.	_____ is an attempt to discover PW by attempting to use every possible PW from a predefined list.							
	a)	Brute-force attack	b)	Dictionary attack	c)	Sniffing attack	d)	Cracking
73.	_____ provides a centralized authentication server to authenticate users to servers and servers to users .							
	a)	Trusted Third Party (TTP)	b)	Trusted Peer Party (TPP)	c)	Trusted Party Model (TPM)	d)	None
74.	_____ is a protocol that <u>produces the same effect as a real signature</u>							
	a)	Digital signatures	b)	Symmetric Key	c)	Asymmetric Key	d)	None
75.	Digital signatures provide the ability to:							
	a)	verify author, date & time of signature	b)	be verified by third parties to resolve disputes	c)	authenticate message contents	d)	All are true
76.	_____ refers to a key storage facility which provides secure storage of keys for future use, e.g.							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
77.	_____ refers to procedures by which keys for notarization or nonrepudiation services can be securely <u>archived</u> .							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
78.	_____ enables parties to update securely their keying material.							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
79.	_____ refers to cryptographic keys which <u>may become lost</u> due to human error, software bugs, or hardware malfunction							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
80.	_____ refers to procedures by which parties are assured of the secure destruction of keys that are no longer needed							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Deletion
81.	_____ refers to the procedure by which keys or pairs of keys of good cryptographic quality are securely and unpredictably generated							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Generation
82.	_____ provide some link between an entity and its uniquely defined keys							
	a)	Entity Registration	b)	Key Replacement	c)	Entity Authentication	d)	Entity Replacement



83.	_____ is to corroborate that an entity is the one claimed							
	a)	Key Replacement	c)	Entity Authentication	d)	Entity Replacement	d)	Entity Registration
84.	_____ is the registration of messages to attest at a later stage its content origin .							
	a)	Notarization	b)	Proof	c)	Nonrepudiation of Reception	d)	none
85.	_____ secret keys and possibly other data are to be kept confidential while transmitted or stored							
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)	Timeliness
86.	_____ is to counter the active threat of unauthorized modification of data items							
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)	Timeliness
87.	_____ is to counter unauthorized duplication of data items.							
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)	Timeliness
88.	_____ is required that the response to a challenge message is prompt and does not allow for playback of some authentic response message by an impersonator.							
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)	Timeliness
89.	_____ It involves storing, safeguarding, and activating keys							
	a)	key management	b)	key distribution	c)	Key Replacement	d)	Key Recovery
90.	_____ refers to the procedures by which keys are securely provided to parties legitimately asking for them .							
	a)	key management	b)	key distribution	c)	Key Replacement	d)	Key Recovery
91.	The _____ private key is used to decrypt cipher text in asymmetric cryptosystem							
	a)	Sender	b)	Receiver	c)	Both A & B	d)	None
92.	The _____ key is used to encrypt plain text in asymmetric cryptosystem							
	a)	public	b)	private	c)	Both A & B	d)	None
93.	_____ is the science of code breaking in a cryptographic schema							
	a)	Cryptography	b)	Cryptanalysis	c)	Cryptology	d)	None
94.	_____ is the study of encryption principles and methods.							
	a)	Cryptography	b)	Cryptanalysis	c)	Cryptology	d)	None
95.	_____ is the field of both cryptography and cryptanalysis.							
	a)	Cryptography	b)	Cryptanalysis	c)	Cryptology	d)	None
96.	In cryptography system, _____ is the original message before being transmitted							
	a)	Cipher text	b)	encrypted	c)	Plain text	d)	None



97.	In Caesar cipher, shifting a letter by +3 positions means that						
	a)	A becomes D, B becomes E, C becomes F, and so on					
	b)	A becomes X, B becomes Y, C becomes Z, and so on					
	c)	A becomes C, B becomes F, C becomes I, and so on					
	d)	A becomes B, B becomes C, C becomes D, and so on					
98.	Type of firewall can decide what services to perform on the user's behalf in accordance with its available knowledge						
	a)	Screening routers	b)	Stateful inspection	c)	Guards	d)
99.	It perform a simple pattern-matching to report situations that corresponding to a known attack type						
	a)	Normal based	b)	Signature-based IDS	c)	Heuristic IDS	d)
100.	It implies overhearing <u>without expending any extra effort</u>						
	a)	Eavesdrop	b)	Wiretap	c)	Spoofing	d)
101.	Triple-DES with two keys id defined by $C= Ek_3(Dk_2(Ek_1(P)))$, if K_1 equal K_3 , the triple-DES can be considered						
	a)	Double DES	b)	Triple DES	c)	Single DES	d)
102.	Which of the following isn't a design goal of firewalls						
	a)	Always invoked	b)	Tamper-proof	c)	Complex code	d)
103.	It tracks the sequence of packets and conditions from one packet to another to thwart an attack						
	a)	Screening routers	b)	Stateful inspection	c)	Application proxies	d)
104.	A sender <u>should not be able falsely to deny later that he/she sent a message</u>						
	a)	Integrity	b)	Confidentiality	c)	Privacy	d)
105.	Physical OS separation approach _____ the resource sharing and _____ the security level						
	a)	Minimizes , Maximizes	b)	Maximizes , Maximizes	c)	Maximizes , Minimizes	d)
106.	Kerberos system is considered _____ party _____ way authentication technique						
	a)	Two , One	b)	Two , Two	c)	Three , Two	d)
107.	<u>Digital signature</u> provides the following characteristics expect _____						
	a)	Message authentication	b)	Receiver authentication	c)	Sender authentication	d)
108.	_____ Is used for managing audit trails and <u>highlighting user violation of policy or normal activity</u>						
	a)	Firewall	b)	IDS	c)	Antivirus	d)
109.	Correction system configuration error is a function of _____						
	a)	Heuristic-based IDS	b)	Personal firewall	c)	Proxy	d)
110.	Denning's IDS model builds heuristic decision model based on tracing _____						
	a)	Actions and it's initiator	b)	Timestamp of each action	c)	Exception condition of each action	d)



111.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
112.	_____ means that some unauthorized party (person, program) has gained access to data							
	a)	Interception	b)	Modification	c)	Interruption	d)	Fabrication
113.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
114.	A virus is designed to hide itself from detection by antivirus software							
	a)	Brain virus	b)	Boot sector virus	c)	Polymorphic virus	d)	Trojan horse
115.	_____ record an intruder's actions and even attempt to trace who the attacker is from actions, packet data, or connection							
	a)	ISS scanner	b)	Nessus	c)	Honey pot	d)	tripwire
116.	Which of the following is/are considered a candidate IDS profile?							
	a)	Logon and session activity	b)	Command or program usage	c)	File access activity	d)	All
117.	_____ used to <i>protect the networks hosts from an outside address</i>							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
118.	Denning's IDS model is example of _____							
	a)	Host based IDS	b)	Signature based IDS	c)	Anomaly based IDS	d)	Hybrid IDS
119.	In which different processes use different physical objects, such as separate printers for output requiring different levels of security							
	a)	Temporal separation	b)	Logical separation	c)	Physical separation	d)	Cryptographic separation
120.	Steganography uses _____ in graphic image <i>to hide important data</i>							
	a)	LSB	b)	HSB	c)	MSB	d)	NSB
121.	_____ Controls access on the basis of packet address or <u>specific transport protocol type</u>							
	a)	Stateful inspection	b)	Screening routers	c)	Application proxies	d)	Guards
122.	A firewall system characterized by being two-headed device is							
	a)	Screening routers	b)	Stateful inspection	c)	Application proxies	d)	Guards
123.	_____ Builds a model of acceptable behavior and flag exceptions to that model							
	a)	Normal based	b)	Signature-based IDS	c)	Heuristic IDS	d)	Honey pot
124.	It attracts the intruder , record his actions, and even attempt to trace who the attacker is from action, packet data, or connections.							
	a)	Heuristic IDS	b)	Honey pot	c)	Normal Based	d)	Signature-based IDS
125.	Intercepting and carrying on a session begun by another entity is known as _____ attack							
	a)	Eavesdrop	b)	Wiretap	c)	Spoofing	d)	Session hijacking



126.	_____ this form of protection <i>limits not just the access to an object but the use made of that object</i> after it has been accessed							
	a)	Share via access limitation	b)	Share all or share nothing	c)	Share Via capabilities	d)	Limit use of an object
127.	_____ is when one host pretends to be another.							
	a)	Eavesdrop	b)	Spoofing	c)	Wiretap	d)	Session hijacking
128.	With this form of protection, the owner of an object declares it to be public or private							
	a)	Share via access limitation	b)	Share all or share nothing	c)	Don't protect model	d)	Isolate mode
129.	_____ in which processes conceal their data and computations in such a way that they are unintelligible to outside processes							
	a)	Temporal Separation	b)	Logical Separation	c)	Physical separation	d)	Cryptographic separation
130.	_____ Change control on the OS instructions, when a computer is started							
	a)	Memory resident virus	b)	Boot sector virus	c)	One time execution	d)	All
131.	When <u>OS runs the applications</u> of different security requirements <i>under illusion that no other processes exist</i>							
	a)	Logical separation	b)	Physical separation	c)	Temporal separation	d)	Space separation
132.	_____ perform simple pattern-matching based on logon and session activity to detect attack							
	a)	Heuristic -based IDS	b)	Personal Firewall	c)	Proxy	d)	Signature-based IDS
133.	Firewall is an executable code runs on a dedicated device to protect inside from outside networks							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
134.	A company wants to block the connection between its LAN and "youtube " , simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
135.	A company wants to set up an online price list so that outsiders can see the products and prices offered. It wants to be sure that no outsider can change the prices or product list, the outsiders can access only the price list, not any of the more sensitive files stored inside.							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
136.	A company wants to allow its employees to use e-mail up to a limit of so many messages or so many characters of e-mail in the last so many days, simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
137.	Which of the following is best description of Stateful inspection firewall working principle?							
	a)	It can see address and services protocol type	b)	It can see full data portion of packet	c)	It can see either address or data	d)	It can see full text of communication



138.	What is true about IDS?				
	a)	it could be stand-alone device attached to the network to monitor traffic throughout that network			
	b)	Honey pot is considered an example for IDS			
	c)	Regarding anomaly-based IDS, admin can mark a flagged behavior as acceptable or not.			
	d)	All			
139.	_____ means intercepting and carrying on a session begun by another entity.				
	a)	Masquerade	b)	Man-in-the-Middle Attack	c) Session hijacking d) Impersonation
140.	Which of the following can describe the function of Firewall system?				
	a)	To encrypt all network traffic to prevent eavesdropping			
	b)	To block all incoming network traffic except for authorized sources			
	c)	To prevent all unauthorized activity of networks' internal users			
	d)	To monitor internal network traffic and detect potential security breaches			
141.	Denning's model is an example for _____				
	a)	heuristic based IDS	b)	signature based IDS	c) Virus Scan software d) Firewall model
142.	Which profile is recommended for auditing?				
	a)	logon and session activity	b)	command or program usage	c) file access activity d) All
143.	When OS runs different process of different security requirements on the same time with shared resources under illusion that no other processes exist				
	a)	Cryptographic separation	b)	Temporal separation	c) logical separation d) Physical separation
144.	if different processes can run concurrently unaware of the presence of each other , and each process has its own address space, files, and other objects. This OS level of sharing is categorized as:				
	a)	Isolation level	b)	Share via capabilities	c) Share via access limitations d) Share all or share nothing
145.	What is true about Kerberos authentication protocol?				
	a)	It is a two-party, two-way authentication protocol			
	b)	It is a two-party, three-way authentication protocol			
	c)	It is a three-party, two-way authentication protocol			
	d)	It is a three-party, three-way authentication protocol			
146.	The ticket issued to the user by Kerberos protocol includes the following data				
	a)	User login details	b)	Time-date stamp	c) User privilege on the accessed application d) All are true
147.	What is a common example of a man-in-the-middle attack				
	a)	Trojan horse	b)	DDoS attack	c) Wi-Fi eavesdropping d) SQL injection



True/False:

1.	<u>Spoofing</u> attack consist of <u>replacing the valid</u> source and/or destination IP address <u>with false one</u>	T
2.	<u>Bug</u> is a <u>mistake in interpreting a requirement</u> or <u>syntax error in a piece of code</u>	T
3.	For exchanging symmetric key among 10 users, we need 100 exchange operation (45)	F
4.	Erasure of program or data or malfunction of an OS file manager is considered fabrication (interruption)	F
5.	<u>Validation error</u> , <u>domain error</u> , and <u>boundary condition violation</u> are types of <u>flaws</u>	T
6.	Erasure of program or data or malfunction of an OS file manager is considered Modification (interruption)	F
7.	<u>Timely response</u> is an expectation of <u>availability</u> security goal.	T
8.	<u>Verification of the user's identity</u> is known as authorization process (authentication)	F
9.	Digital signatures provide the ability to verify both owner and content of message	T
10.	The plain text must be broken into a block size of 128bit as an input of <u>Triple DES</u> (64-bit)	F
11.	<u>Polymorphic virus</u> is a form of virus explicitly designed to <u>hide itself from detection</u>	T
12.	A <u>boot sector</u> virus changes control on the OS instructions, when a computer <u>is started</u>	T
13.	Playfair cipher rearranges the order of words in a message (letters)	F
14.	Shift cipher applies a <u>mathematical formula</u> to each letter in a message that replaces each letter with a numerical value. (Vigenère)	F
15.	<u>Flooding</u> is an attack of <u>availability</u>	T
16.	Authorization is mechanism used to confirm a user's identity (authentication)	F
17.	Fingerprint is used to verify the user as an example for something <u>user has</u> (ATM card)	F
18.	Trapdoor is a malicious software runs when a computer is started, it Changes control on the OS instructions (boot sector virus)	F
19.	Program controls must include <u>peer review steps</u>	T
20.	In context of Program controls, <u>proof of program correctness</u> is done based on <u>check list</u>	T
21.	<u>Code flaws</u> include validation error or domain error	T
22.	Heuristic IDS performs simple pattern-matching to report situation that corresponding to a known attack type (signature-based)	F
23.	<u>Eavesdrop</u> Implies overhearing <u>without expending any extra effort</u>	T



PRACTICAL BANK

Choose the correct answer:

1.	Ensuring that computer assets are accessed only by authorized parties and authorized parties actually access the required assets is a goal of _____.			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
2.	_____ is Ensuring that the data is accurate, consistent, and not tampered with .			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
3.	Using systems like Git to track and prevent unapproved modifications in codebases achieves _____ of the data			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
4.	_____ Is Ensuring that the systems and the data are accessible when needed .			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
5.	Service is <u>denied to unauthorized users</u> is an example of data availability (confidentiality)			
	a) TRUE	b) FALSE	c)	d)
6.	When an asset of the system is destroyed or becomes unusable , this is called _____ and this is an attack on the _____			
	a) Interruption, availability	b) Interception, availability	c) modification, availability	d) modification, integrity
7.	An unauthorized party only gains access to an asset this is called _____ and this is an attack on the _____.			
	a) Interruption, availability	b) Interception, confidentiality	c) modification, availability	d) modification, integrity
8.	An unauthorized party not only gains access to an asset but tamper with an asset this is called _____ and this is an attack on the _____.			
	a) Interruption, availability	b) Interception, confidentiality	c) modification, availability	d) modification, integrity
9.	“ Who are you ”? is an _____ question while “ What are you allowed to do ?” is an _____ question			
	a) Authentication, Authorization	b) Authorization, Authentication	c) Authentication, Authentication	d) Authorization, Authorization
10.	_____ is a network security device that <i>monitors incoming and outgoing network traffic</i> and decides whether to allow or block specific traffic .			
	a) Switch	b) Firewall	c) Router	d) Hub
11.	In Huawei eNSP, which command is used to add an interface to a security zone ?			
	a) add interface			
	b) interface add			
	c) firewall zone add			
	d) firewall zone [zone_name] add interface			



12.	Which of the following is NOT a default security zone in Huawei firewalls?			
	a) Trust	b) DMZ	c) Untrust	d) ISP
13.	___ ensures high availability of the default gateway by allowing multiple routers to form a virtual gateway .			
	a) VRRP	b) ISP	c) Virtual switches	d) All of the above
14.	Which command is used to enable VRRP on an interface in Huawei routers (VRP CLI)?			
	a) vrrp enable			
	b) vrrp vrid <id> virtual-ip <ip>			
	c) vrrp activate			
	d) ip vrrp enable			
15.	Which command is used to enter the interface where VRRP will be configured?			
	a) interface vrrp 1			
	b) vrrp interface GigabitEthernet0/0/1			
	c) interface GigabitEthernet0/0/1			
	d) int vrrp 1			
16.	Which command is used to define the source security zone in a policy rule?			
	a) source-zone trust	b) src-zone trust	c) zone source trust	d) from trust
17.	Which of the following is not a valid action in a security policy rule?			
	a) permit	b) deny	c) drop	d) reject
18.	Which command displays detailed information about VRRP configuration and status on a Huawei device?			
	a) show vrrp status	b) display vrrp	c) display standby all	d) display ip vrrp
19.	What does NAT stand for?			
	a) Network Application Tool			
	b) Network Access Terminal			
	c) Network Address Translation			
	d) Network Authentication Type			
20.	What is the correct command to enter system mode on a Huawei device?			
	a) enable system	b) system-mode	c) system-view	d) enter global-mode
21.	What is the function of the quit command in Huawei CLI?			
	a) Exits the CLI and shuts down the device			
	b) Exits the current configuration mode and returns to the previous level			
	c) Saves the configuration and exits system-view mode			
	d) Logs out the user from the device			



22.	Which command is used to <u>define a named security policy rule</u> in Huawei firewalls?							
	a)	security-policy name [rule-name]						
	b)	rule name [rule-name]						
	c)	security-policy rule name [rule-name]						
	d)	rule [rule-name] security-policy						
23.	What is the primary purpose of NAT ?							
	a)	Encrypt traffic between routers						
	b)	Provide DNS translation						
	c)	Translate private IP addresses to public IP addresses						
	d)	Assign static routes automatically						
24.	Which of the following is a correct syntax for defining a NAT address group?							
	a)	ip nat pool 1 range 202.100.1.1 202.100.1.5						
	b)	nat address-group 1 202.100.1.1 202.100.1.5						
	c)	nat ip-pool 1 from 202.100.1.1 to 202.100.1.5						
	d)	address-group nat 1 202.100.1.1 to 202.100.1.5						
25.	Which command is most commonly used to <u>test basic IP connectivity between two devices</u> ?							
	a)	display ip routing-table	b)	ping	c)	tracert	d)	telnet
26.	_____ command shows active session entries handled by the firewall , including source/destination IPs, ports, protocol, and interface.							
	a)	display firewall policy						
	b)	display firewall state-table						
	c)	show firewall connections						
	d)	display firewall session table						
27.	What does GRE stand for?							
	a)	Global Routing Encryption						
	b)	General Remote Exchange						
	c)	Generic Routing Encapsulation						
	d)	General Route Establishment						
28.	What is the primary function of GRE?							
	a)	Encrypt data between routers						
	b)	Compress IP packets						
	c)	Encapsulate one network protocol within another						
	d)	Filter traffic at Layer 4						



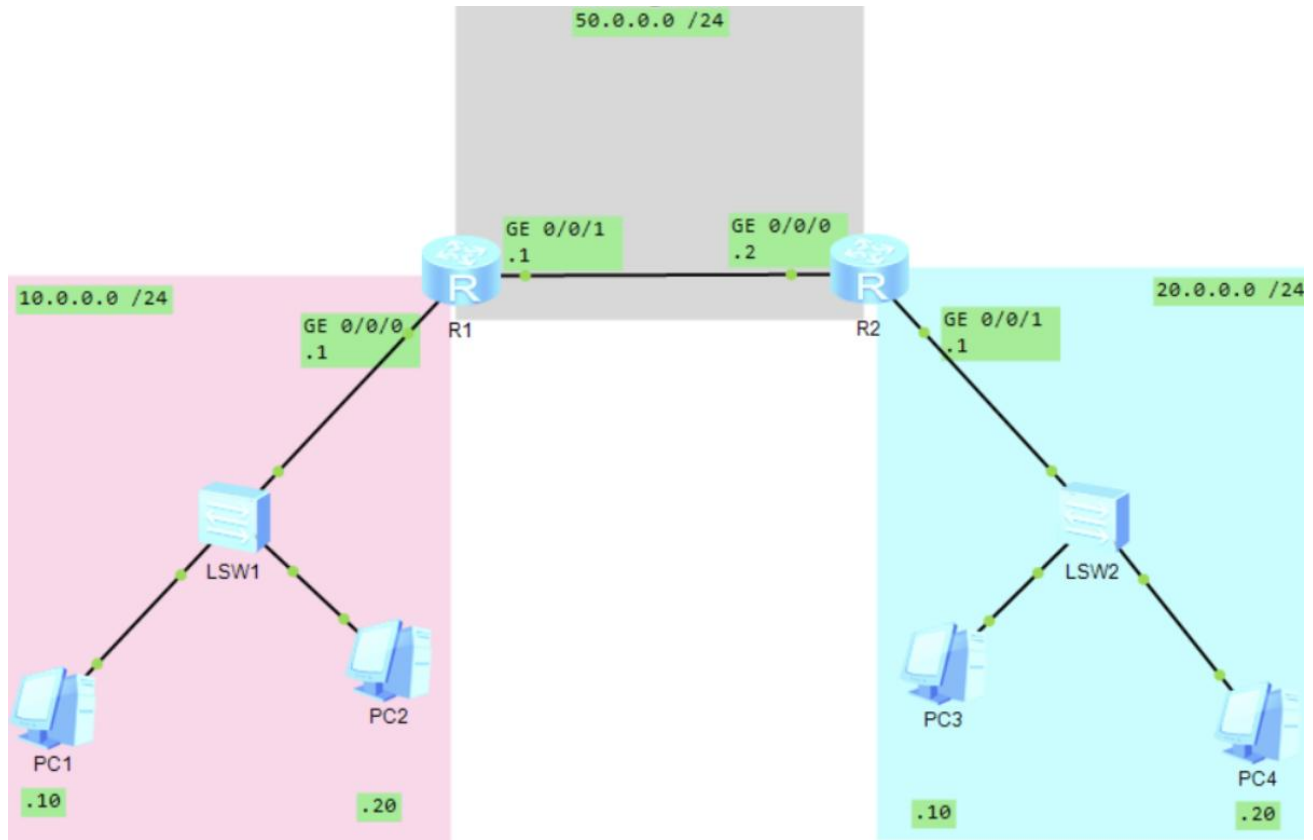
29.	What must be specified when configuring a GRE tunnel?						
	a)	Only the destination port					
	b)	ACL and NAT rules					
	c)	Tunnel source and destination IP addresses					
d)	DNS and hostname						
30.	What is the correct command to <u>configure a static route</u> in Huawei VRP?						
	a)	ip route-add 192.168.2.0 255.255.255.0 10.0.0.1					
	b)	route static 192.168.2.0 255.255.255.0 10.0.0.1					
	c)	ip route-static 192.168.2.0 255.255.255.0 10.0.0.1					
d)	set route 192.168.2.0 255.255.255.0 via 10.0.0.1						
31.	Which mode must you be in to <u>configure a static route</u> ?						
	a)	Interface mode	b)	User view	c)	System view	d)
32.	Which command <u>starts the configuration of a GRE tunnel interface</u> in Huawei VRP?						
	a)	interface gre0/0/0					
	b)	interface Tunnel0/0/0					
	c)	gre tunnel 0					
d)	interface VPN-GRE0						
33.	What is the correct command to <u>set the GRE tunnel mode</u> ?						
	a)	encapsulation gre					
	b)	set tunnel gre					
	c)	tunnel-protocol gre					
d)	gre-enable						
34.	Which command defines where the tunnel is going ?						
	a)	destination-tunnel 192.168.2.1					
	b)	destination 192.168.2.1					
	c)	peer-ip 192.168.2.1					
d)	set next-hop 192.168.2.1						
35.	What is the correct command to <u>assign an IP to a GRE tunnel interface</u> ?						
	a)	ip address 10.0.0.1/30					
	b)	set ip tunnel 10.0.0.1 255.255.255.252					
	c)	ip address 10.0.0.1 255.255.255.252					
d)	tunnel-ip 10.0.0.1						



36.	What will happen <u>if the destination IP in a GRE tunnel is incorrect or unreachable?</u>			
	a)	The tunnel still works, but slowly		
	b)	Traffic will reroute through another tunnel		
	c)	The tunnel will remain down and won't forward traffic		
	d)	The router will reboot automatically		
37.	What command is used to <u>assign an IP address to the interface?</u>			
	a)	ip address add		
	b)	set ip GigabitEthernet		
	c)	ip address 192.168.3.1 255.255.255.0		
	d)	address 192.168.3.1 mask 24		
38.	What command is used to <u>block a specific group of IP addresses from accessing a certain resource?</u>			
	a)	ip address-set object server-deny type list		
	b)	ip address-set server-deny type object		
	c)	ip address-set deny object		
	d)	ip set server-deny object type address		
39.	Which command correctly creates a time-range called work-hours that applies every Monday to Friday from 08:00 to 9:00?			
	a)	time-range work-hours period-range Monday to Friday 08:00 to 9:00		
	b)	time-range work-hours cycle Monday-Friday 08:00-17:0		
	c)	set time-range work-hours recurring Monday to Friday 08:00 17:00		
	d)	ip set server-deny object type address		
40.	Which command <u>enables the firewall function</u> on a Huawei router in eNSP?			
	a)	ip firewall enable		
	b)	firewall enable		
	c)	security firewall enable		
	d)	firewall startup		
41.	Which command is used to <u>translate internal server IPs for public access?</u>			
	a)	nat static	b) nat server	c) nat overload
42.	_____ is a routing protocol used to <u>determine the best path for data to travel through a network.</u>			
	a)	http	b) rip	c) smtp
43.	To enable dynamic routing on eNSP routers, you need to enable _____.			
	a)	rip	b) acl	c) both



Use the following topology to answer the following questions:



44. To enable static routing (NAT – network address translation) for **R1**, we use the command
syntax: ip route-static ip-address { mask | mask-length } { nexthop-address }
 هنا الامر المظبوط هي يبقى الي اوله ip route-static وبعدين بتكتب عنوان ال r1 الي هيكون 0.0.0.0 والماسك بتاعه 0 وبتكتب بعده المكان الي انت عايز توصل الراوتر بيه الي هو R2 الي عنوانه 50.0.0.2
- | | |
|-----------|--|
| a) | static routing ip 0.0.0.0 0 |
| b) | ip route-static 0.0.0.0 0 50.0.0.2 24 |
| c) | ip route-static 0.0.0.0 0 10.0.0.1 24 |
| d) | static routing ip 0.0.0.0 0 10.0.0.1 24 |
45. To enable static routing (NAT – network address translation) for **R2**, we use the command:
- | | |
|-----------|---|
| a) | static routing ip 0.0.0.0 0 |
| b) | ip route-static 0.0.0.0 0 50.0.0.1 |
| c) | ip route-static 0.0.0.0 0 20.0.0.1 |
| d) | static routing ip 0.0.0.0 0 20.0.0.1 24 |
46. To assign public address for **PC1**, we use the command:
- | | |
|-----------|---|
| a) | nat static global 50.0.0.10 inside 10.0.0.0 |
| b) | nat static global 50.0.0.10 inside 10.0.0.10 |
| c) | nat static 50.0.0.10 10.0.0.10 |
| d) | nat static 10.0.0.10 50.0.0.10 |



47.	You need to assign IP address for each interface in a router or firewall using the command:							
	a)	ip address 200.0.0.1 24						
	b)	p address 200.0.0.0 24						
	c)	both						
	d)	none						
48.	To display NAT information for a specific PC , we use the command:							
	a)	display nat session all						
	b)	display nat static inside ip_for_pc						
	c)	display ip						
	d)	display net						
49.	To make a pool of addresses for dynamic routing , we use the command:							
	a)	nat address-group 1 50.0.0.50 50.0.0.55						
	b)	nat address-group 50.0.0.50 50.0.0.55						
	c)	routing address-group 50.0.0.50 50.0.0.55						
	d)	static address-group 50.0.0.50 50.0.0.55						
50.	To activate rule number 5 from ACL , we use the command:							
	a)	rule 5 permit source any						
	b)	acl permit						
	c)	route acl						
	d)	rip acl						
51.	If we have an address pool # 2, we can assign the outbound for dynamic routing using the command:							
	a)	nat outbound 2000 address-group 2 no-pat						
	b)	assign #2 nat						
	c)	address-group 2 dynamic						
	d)	nat inbound 3000						
52.	A firewall is a _____ area filter.							
	a)	local	b)	global	c)	logical	d)	all
53.	By default, the firewall allows all traffic from a lower security zone to a higher security zone? (blocks)							
	a)	TRUE	b)	FALSE	c)		d)	
54.	NAT could mask the host port number with another port number							
	a)	TRUE	b)	FALSE	c)		d)	
55.	_____ is a defined set of security rules							
	a)	Security Zone	b)	Security Policy	c)	rule name	d)	None



56.	To define <u>Backup</u> group in vrrp we write <i>standby</i> command in code of creating group?							
	a)	TRUE	b)	FALSE	c)		d)	
57.	passing data between master and backup group made by using GRE protocol (HRP)							
	a)	TRUE	b)	FALSE	c)		d)	
58.	VRRP stands for _____.							
	a)	Vehicle Router Redundancy Protocol						
	b)	Virtual Router Redundancy Protocol						
	c)	Virtual Router Retransformation Protocol						
	d)	All						
59.	_____ command shows active session entries handled by the firewall , including source/destination IPs, ports, protocol, and interface.							
	a)	show firewall session table						
	b)	display firewall session table						
	c)	Both						
	d)	None						
60.	you define the nat policy before you define the nat address group (after)							
	a)	TRUE	b)	FALSE	c)		d)	

